

[CH 11] Broadening The Appeal

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- Like most technologies, the computer started out imperfect, difficult to master and was only used by enthusiastic early adopters
- In the early 1980s, the GUI increased ease of use, more software and services came out, CD-ROMs brought large amounts of information, chatting and email enabled users to reach out to each other
- This made able to use the computer, but it also made them **want** to use it

Maturing of the PC software industry

- 1980, microsoft had 38 employees and sales of \$8 million/year
- 1990 it had 5600 employees and sales of 1.8 billion/year
- 24 August 1995 they launched Windows 95
 - Their most important software project to date
 - The launch publicity cost \$200 million
- Their rise was based on software, not hardware
- The PC industry developed in 2 phases:
 - The gold rush era 1975-1982
 - Low barriers to entry
 - Thousands of new entrants (undercapitalised 2 or 3-person start-ups)
 - Second phase 1983 onwards
 - Standardization around the IBM PC
 - Period of consolidation
 - Barriers of entry heightened, many early firms disappeared
 - A few firms emerged as global players
- The PC industry and the large software firms were disconnected
- Large software companies, used to developing large reliable programs were irrelevant when developing programs for tiny PC memories
- Culture was also different: PC software companies sold through mail order / retail outlets and the atmosphere was informal
- PC software would need to evolve its own style of marketing: magazines, free demo diskettes, point-of-sale materials, exhibitions
- Marketing usually cost twice as much as development
- By 1983, 15 companies had 2/3 of the market
- 3 reasons for the high barriers of entry:
 - PC's were rapidly increasing in performance, capable of running the same software as small mainframes. This meant that it needed the same kind of resources to develop for it.
 - The need for know-how. The information about how to design good, appealing applications for PC was not in literature or in comp. sci. classes. It was locked in companies
 - Distribution channels. There were tens of times more products on the market than stores could sell => they had to compete for shelf space through marketing => need for large injections of capital
- Major firms like VisiCorp, MicroPro and Ashton-Tate were not immune
- A software "hit" would keep a company going for years, but once its success faded, so would the company
 - VisiCorp (creators of VisiCalc spreadsheet) went from \$40 million in 1983 to not existing in 1985, wiped out by Lotus 1-2-3 spreadsheet
- Lotus 1-2-3 was developed by Mitch Kapor, who had experience creating packages for VisiCalc
 - He used the VisiCalc royalties and the money he got from selling the rights, along with venture capital to built his own competitor to VisiCalc: Lotus 1-2-3
 - It Cost \$1 million to develop but 2.5 million to market
 - It became market leader in the first 18 months
- Microsoft relied on income from already successful products, instead of venture capital
 - The success of MS-DOS carried it through many unsuccessful projects

Graphical User Interface

- The PC had to become more user friendly to attract more users
 - This was achieved through Macintosh computers or through Microsoft Windows
 - Underlying both was the GUI concept
 - The existing software used a command line interface
 - Commands had to be letter-perfect
 - It was difficult and irritating to work with for ordinary people
 - Technical people liked its intricacies, but that was not the majority of users
 - Most of the technologies required for this already existed
 - Doug Engelbart's Human Factors Research Center at the SRI (founded 1963), with funding from ARPA, developed the mouse
 - Devenber 1968 a prototype electronic office was demonstrated by Engelbart's team
 - At the University of Utah's Computer Science Laboratory, under the leadership of David Evans and Ivan Sutherland, computer graphics innovations were taking place
 - Alan Kay developed a device that would fulfill the personal-information needs of an individual and called it the Dynabook
 - The Dynabook and Engelbart's electronic office led to the modern GUI
 - They were held back by the lack of compact cost-effective technology
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- Xerox Corporation set out to develop the technology for "the office of the future"
 - It hired former ARPA Information Processing Techniques Office head Robert Taylor to work at their new Palo Alto Research Center (PARC) in Silicon Valley
 - The result of Taylor's work was the Alto desktop computer
 - During this, Xerox PARC developed the GUI, which became the preferred style of the 1980s
 - Steve Jobs, impressed by the Alto, recruited Larry Tesler from Xerox to develop a new apple product called the Lisa
 - The Lisa took 3 years to develop and when it finally came out at \$16,995, it was too expensive
 - Lisa was a failure and left apple exposed, with its last successful product (Apple II) already getting old
 - What saved Apple was the Macintosh
 - Coming in at \$2500 in 1984, the initial sales were disappointing
 - The concept of the PC as a consumer appliance was not practical
 - The average person couldn't do enough with a computer to justify the price
 - It was relaunched, targetting the business machine market. It became the machine of choice in the publishing and media industries. In more conservative industries, IBM-compatible PC's were still favoured
 - Microsoft developed software applications for the Macintosh market, isolated from the highly competitive IBM-compatible market. It was later able to port these to IBM-compatible systems

Microsoft Windows

- The launch of the macintosh made other computers look dull, old-fashioned and lackluster